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## 04 How to interpret this report

Our methodology, what affects results, and best practices

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Time Crystals: From a No-Go Theorem to Driven Phases of Matter Soumyabroto Majumder1, Rahul Chandra2, Analabha Roy1, and Mahbub Rahaman3,* 1Department of Physics, The University of Burdwan, Burdwan 713104, India 2Sankara Nethralaya Elite School of Optometry, DJ/16, Action Area 1D, New Town, Rajarhat, Kolkata 700156, India 3Harish Chandra Research Institute, A CI of Homi Bhabha National Institute, Chhatnag Road, Jhunsi, Prayagraj, Uttar Pradesh 211019, India *Contact author: mahbub.phys@gmail.com Abstract Spontaneous symmetry breaking is fundamental to the emergence of spatial order in condensed matter and occurs when a ground or thermal state possesses less symmetry than its governing Hamiltonian.
Wilczek proposed an analogous breaking of time translation symmetry in which a quantum system in its lowest energy state would exhibit persistent periodic motion.
Although this equilibrium proposal was subsequently challenged, the concept of time crystalline order was reformulated in nonequilibrium systems.
In periodically driven systems, continuous time translation symmetry is reduced to a discrete symmetry, whose spontaneous breaking can give rise to a discrete time crystal (DTC) characterized by robust oscillations with a period equal to an integer multiple of the driving period.
Because generic driven systems absorb energy and eventually heat toward infinite temperature, the emergence of such order requires mechanisms, including many-body localization, prethermalization, or integrability that suppress heating.
We review the Floquet algebraic origin of long lived oscillations and examine their dependence on driving protocols, initial states, and interactions, while highlighting theoretical and experimental advances and the roles of symmetry, dimensionality, and dissipation in stabilizing DTCs.
Keywords: time crystals; Floquet systems; integrability; dynamical symmetries I.
INTRODUCTION ferromagnet breaks spin rotation Symmetry is a central organizing principle symmetry, while a crystal breaks continuous spatial translations down to a of nature.
In condensed matter, the spontaneous breaking of symmetry discrete lattice subgroup.
In both cases, the ground state or thermal state exhibits less underlies the emergence of order.
A 1 symmetry than the Hamiltonian from breaking, must be identified through which it arises.
The search for new phases correlation functions.
Watanabe and of matter, and for the symmetries they Oshikawa therefore characterized TTSB break, is central to the development of using the correlator of an extensive condensed matter physics and to observable $O = \int d^d x\, O(\mathbf{x})$, technological progress.
Since relativity 1 − places space and time on an equal footing, $f(t) = \lim \langle e^{iHt} O e^{iHt} O \rangle$ , (1) $V \to \infty$ V2 it is natural to ask whether time-translation and showed that f(t) is time independent in symmetry can be broken in the same way.
the ground state and in Gibbs states of In 2012, Wilczek and Shapere posed this local Hamiltonians.
The Gibbs state result question for quantum (Wilczek, 2012) and was later established rigorously (Watanabe classical (Shapere & Wilczek, 2012) et al., 2020).
Time crystals are therefore systems, respectively, and introduced the forbidden in equilibrium.
term time crystal.
Their proposal The idea of TTSB was revived by envisioned a quantum system in its lowest considering quantum systems driven far energy state undergoing perpetual periodic from equilibrium.
In a periodically driven motion, such as a rotating density many-body quantum system, the modulation of interacting charged bosons Hamiltonian satisfies H(t + T) = H(t), on a ring threaded by magnetic flux.
and only the discrete time-translation Although this motion would not permit symmetry $t \to t + T$ remains.
If the state energy extraction, it appeared inconsistent and the corresponding observables respond with the conventional understanding that with a period mT, where $m \geq 1$ is an equilibrium states are stationary.
The integer, this discrete symmetry is proposal was soon challenged: Bruno spontaneously broken.
This is the essence (2013) showed that the persistent motion of a discrete time crystal (DTC) (Else et envisaged in Wilczek's ring model cannot al., 2016; Khemani et al., 2016; Sacha, occur in a ground state, while Watanabe 2015; von Keyserlingk et al., 2016; Yao et and Oshikawa (2015) established a general al., 2017).
The response must also be no-go theorem.

Since every eigenstate and robust against perturbations, accompanied Gibbs state is stationary at the level of by spatial long-range order, and persist in one-point observables, time-translation the thermodynamic limit.
The main symmetry breaking (TTSB), like obstacle is that generic interacting driven conventional spontaneous symmetry systems absorb energy and heat to a 2 featureless infinite-temperature state.
The many body localization is unavailable and first proposals avoided this through robustness must come from other disorder-induced many-body localization, mechanisms.
We also discuss the role of while prethermalization at high driving integrability, which suppresses heating and frequency and long-range interactions can sustain persistent oscillations, and ask were later identified as disorder-free routes when these oscillations survive (Else et al., 2017; Pizzi et al., 2021; integrability breaking perturbations and so Russomanno et al., 2017).
Soon after these constitute genuine time crystalline order.
proposals, signatures of period doubling Section II introduces the Floquet were reported in trapped ions (Zhang et al., formalism, the criteria for time crystalline 2017) and in dense ensembles of nitrogen-order, and the main stabilization vacancy (NV) centers (Choi et al., 2017).
mechanisms. Section III examines These were followed by observations in integrability and the dynamical NMR systems (Pal et al., 2018; Rovny et symmetries that underlie persistent al., 2018), nuclear-spin registers (Randall oscillations.
Section IV reviews et al., 2021), prethermal trapped-ion chains experimental realizations, and Sec.
V (Kyprianidis et al., 2021), and discusses open problems.
superconducting quantum processors (Mi II.
DISCRETE TIME CRYSTALS IN et al., 2022).
Comprehensive reviews of FLOQUET SYSTEMS the field are available (Else et al., 2020; Khemani et al., 2019; Sacha & A.
Floquet formalism and criteria for Zakrzewski, 2018; Zaletel et al., 2023), time-crystalline order together with a monograph (Sacha, 2020).
With the no-go theorem, the search for In this short review, we focus on the time crystals shifted to nonequilibrium Floquet algebraic origin of long-lived systems.
The first proposals were for oscillations and their dependence on periodically driven (Floquet) systems, driving protocols, initial states, and which retain only a discrete time interactions.
We highlight theoretical and translation symmetry.
To break this experimental advances and the roles of symmetry, the system must respond with a symmetry, dimensionality, and dissipation period mT that is an integer multiple m ≥ in stabilizing DTCs.
We restrict our 1 of the drive period T.
To understand how attention to disorder-free spin-1/2 systems, such a response can arise, we first review both interacting and integrable, where the Floquet formalism.
3 For a Hamiltonian with period T = 2π/ωD, periodic but repeats only after m ≥ 1 H(t + T) = H(t), the wavefunction |ψ(t)⟩ periods, obeys the Schrodinger equation i∂ |ψ ⟩ t (t) f((n + 1)T)≠ f(nT), (3) = H(t)|ψ(t)⟩ (we set $\hbar$ = 1).
The f((n + m)T)= f(nT).
stroboscopic evolution, which determines This manifests as a subharmonic response, the observables at times t = nT, is which can be readily observed as a peak at generated by the Floquet operator frequency ωD/m in the Fourier spectrum of f(nT).
Equivalently, the Floquet UF ≡ U(T,0) T eigenstates are long range correlated cat = Texp −i H(t) dt 0 (2) states.
Their quasienergies are organized in − iHFT =e , multiplets split by ωD/m, which is the ⟩ n |ψ(nT) =UF |ψ(0)⟩, Floquet analogue of the correlator criterion where T denotes time ordering and HF is in Eq.
(1) (Else et al., 2016; Khemani et the Floquet Hamiltonian.
By Floquet's al., 2017; von Keyserlingk et al., 2016).
theorem, the solutions of the Schrodinger A subharmonic response alone, however, − μ equation can be written as |ψ (t)⟩ = e i αt|α does not guarantee a discrete time crystal.
ϕ (t)⟩, where |ϕ (t + T)⟩ = |ϕ (t)⟩ and the The response must emerge from the α α α quasienergies μ are defined modulo ωD.
collective dynamics of the interacting α The Floquet eigenstates |ϕ (0)⟩ are driven system.
Non-interacting spins α subjected to perfect π pulses flip every invariant under UF up to a phase.
They therefore maintain the discrete time period and trivially show period doubling.
translation symmetry t→t + T, and play A spin rotation error εr, however, shifts the the role that energy eigenstates and Gibbs

response frequency from π/T to π(1−εr)/T, states play in equilibrium.
Any breaking of so the period is not locked.
To label a this symmetry must therefore be system a discrete time crystal, it must spontaneous.
therefore satisfy the following criteria (Else et al., 2016; Khemani et al., 2016; Discrete time translation symmetry is Yao et al., 2017): spontaneously broken if, for generic short range correlated initial states |ψ(0)⟩ (for a) The system should exhibit instance product states), the stroboscopic spontaneous breaking of discrete time translation symmetry: for expectation value f(nT) = ⟨ψ(nT)|O|ψ generic short-range correlated (nT)⟩ of some local observable O is not T-4 initial states, the stroboscopic (ETH) (D'Alessio et al., 2016; Deutsch, expectation value of a local 1991; Srednicki, 1994).
Local observables observable should satisfy Eq.
(3), then relax to their infinite-temperature even though the Hamiltonian is values, which destroys the subharmonic periodic, H(t) = H(t + T).
response (D'Alessio & Rigol, 2014; Lazarides et al., 2014a).
A central b) The response should be rigid: the challenge is therefore to identify subharmonic oscillations should mechanisms that suppress heating while remain locked at period mT over a stabilizing the subharmonic dynamics.
One finite range of parameters, widely studied approach is to localize the including imperfect pulses, without many-body system throughout the drive fine tuning the system parameters.
cycle. Other approaches slow down or c) The oscillating order parameter prevent heating without localization.
should be long-range correlated in Several theoretically and experimentally space, and the oscillations should established mechanisms can provide such persist indefinitely in the localization or otherwise suppress heating.
thermodynamic limit.
The limits The most widely used approach is to must be taken in the order limnT →∞ introduce strong disorder, which can limN , since in any finite system →∞ produce many-body localization (MBL) tunnelling between the cat state (Abanin et al., 2019; Basko et al., 2006; partners eventually destroys the Pal & Huse, 2010).
An MBL system oscillation.
possesses an extensive set of quasi-local Therefore, the dynamics of a periodically integrals of motion (Huse et al., 2014; driven interacting system must support a Serbyn et al., 2013), and MBL has been robust subharmonic response that persists proven in one-dimensional spin chains in the thermodynamic limit.
In generic under an assumption on level statistics driven systems, however, energy (Imbrie, 2016).
MBL persists under absorption from the drive leads to heating periodic driving (Lazarides et al., 2015; toward infinite temperature.
Because a Ponte et al., 2015).
Its quasi-local integrals periodic drive conserves no energy, the of motion inhibit energy absorption from Floquet eigenstates of a generic interacting the drive and thus prevent heating to system satisfy a Floquet version of the infinite temperature.
MBL is stable against eigenstate thermalization hypothesis weak perturbations within the isolated 5 system.
It is, however, destroyed by exponentially long heating time rather than coupling to an environment, and its strictly forever.
stability in higher dimensions, with long-Additionally, precise engineering of the range interactions, and asymptotically drive protocol can strongly suppress even in one dimension remains under thermalization.
At special drive active debate (De Roeck & Huveneers, parameters, two-tone drive protocols 2017; Morningstar et al., 2022; Šuntajs et manifest exactly flat Floquet bands and al., 2020).
thereby suppress heating (Banerjee et al., A second approach is to use a clean, 2025).
More generally, the leading-order disorder-free system driven at high Floquet Hamiltonian can acquire emergent frequency.
When the drive frequency is conservation laws at special drive large compared with the local energy scale amplitudes and frequencies, which Jloc, heating is exponentially slow, with a constrains the dynamics and produces − ω rate bounded by e c D/Jloc (Abanin et al., long-lived nonthermal behavior.
2017; Kuwahara et al., 2016; Mori et al., A further approach is to consider an 2016).
The system then enters a prethermal integrable system whose dynamics are regime in which its dynamics are constrained by an extensive set of governed, for long but finite times, by an conserved quantities.
Such a system effective time-independent Hamiltonian relaxes to a generalized Gibbs ensemble before eventual heating occurs.
By the rather than to infinite temperature.
Under Watanabe-Oshikawa theorem, such a static appropriate conditions, integrability can effective Hamiltonian alone cannot

host a then support persistent oscillations.
These time crystal.
For a drive containing a near-arise when the dynamics admits a π global spin flip P, however, the Floquet dynamical symmetry, an operator A − θ operator takes the form U ≈ P e iDT, † i F satisfying UFA UF = e A, for which θ = π where the effective Hamiltonian D has an gives period doubling.
In isolated emergent Z2 symmetry, D,P = 0 (Else et Hamiltonian systems, the analogous al., 2017).
If the prethermal state of D condition H,A = ωA sustains undamped spontaneously breaks this symmetry, the oscillations (Buča et al., 2019; Medenjak system shows period doubling.
This et al., 2020).
For the kicked Ising chain of requires a finite-temperature ordered z Sec.
II B at perfect π pulses, A = σj phase, and the oscillation persists up to the satisfies this condition with θ = π on every 6 site.
Each of the mechanisms above can B.
Drive protocols and stabilization then be viewed as a way of ensuring that a mechanisms dressed version of z survives pulse errors.
σj The canonical protocol for realizing a We return to dynamical symmetries in Sec.
discrete time crystal is based on a binary III A.
Floquet drive, shown schematically in Fig.
Section II B introduces the generic drive 1.
During each period, the system evolves protocol for a discrete time crystal, shown under two Hamiltonians, H1 and H2, for schematically in Fig.
1, and reviews durations T1 and T2 = T−T1, respectively.
established results on the mechanisms that Defining produce robust discrete time crystalline − − U = e iH1T1, U = e iH2T2, (4) order.
1 2 the one-period Floquet operator is − − iH2T2 iH1T1 UF = U2U1 = e e .
(5) The first step is chosen to implement an approximately global spin flip, while the second is generated by an interacting local or quasi-local Hamiltonian.
For a spin-1/2 chain, a representative choice is (Yao et Figure 1.
Schematic of the discrete time al., 2017) crystal drive protocol.
The system evolves x H = g σi , under two Hamiltonians, H1 and H2, for 1 i (6) durations T1 and T2 = T−T1, respectively.
z z z H = Jiσi σi+1 + hiσi , π 2 The first step, H = ∑ x, i 1 2T1(1−εr) j σj where J and h denote random couplings implements an approximately global spin i i and longitudinal fields.
In the limit T1→0 flip with spin-rotation error εr, while the second is generated by an interacting local with gT1 fixed, the first step becomes an or quasi-local Hamiltonian.
Over a finite instantaneous kick, and the protocol is known as the kicked Ising chain.
range of εr, the resulting stroboscopic magnetization exhibits a subharmonic Experimentally, the global spin flip is implemented by a nominal pulse, which response locked at period 2T.
π is generally imperfect.
We parametrize the pulse imperfection by 7 π of interactions.
In a genuine discrete time g = (1−εr), (7) 2T1 crystalline phase, the response must so that U1 rotates every spin about the x remain stable against perturbations and axis by the angle π(1−εr).
The spin persist indefinitely in the thermodynamic rotation error εr vanishes for ideal pulses.
limit. For an ideal pulse, εr = 0, the first step The defining feature of the discrete time N x crystalline phase is that the subharmonic reduces to U1 = (−i) P with P = ∏i σi , and flips every spin, response remains stable over a finite interval of εr, rather than occurring only at † z z U1σi U1 = −σi , the fine-tuned value εr = 0.
This (8) † z = − σz .
robustness arises from the interacting U1 σi U1 i i i dynamics during the T2 interval, provided The stroboscopic magnetization is that heating is suppressed and the time 1 n crystalline order is preserved.
Fig. 2 = ⟨ ψ | U† M(nT) N 0 F i (9) illustrates this at εr = 0.1.
In the non- σz Un|ψ ⟩.

i F 0 interacting limit, Ji = 0 (with hi = 0), each Since H commutes with every σz, at ε = 2 i r spin simply rotates by π(1−εr) per period, † z z 0 we have UFσi UF = −σi .
This is precisely and the Fourier peak of the magnetization the dynamical symmetry with θ = π shifts from ω = π/T to ω = π(1−εr)/T.
In introduced in Sec.
II A and discussed the presence of interactions, Ji ≥ 0, the further in Sec.
III A.
The magnetization peak remains sharply locked at ω = π/T.
therefore alternates between successive The interaction term thus counteracts the periods, phase error introduced by the finite spin-n M(nT) =(−1) M(0), rotation error εr.
It is therefore crucial for (10) M((n + 2)T=) M(nT).
both the emergence and the stabilization of Thus, the observable responds with period the subharmonic response (Else et al., 2T, rather than with the drive period T, 2016; Khemani et al., 2016; Yao et al., confirming the subharmonic response 2017).
associated with discrete time translation symmetry breaking.
At this fine tuned point, however, the period doubling is trivial, since it holds equally in the absence 8 small in the system size (Else et al., 2016; Khemani et al., 2016; von Keyserlingk et al., 2016).
This is the many-body-localized DTC.
Its rigidity has been established theoretically (Else et al., 2016; Khemani et al., 2016; von Keyserlingk et al., 2016; von Keyserlingk & Sondhi, 2016; Yao et al., 2017), and signatures of it have been Figure 2.
(a) Stroboscopic magnetization observed experimentally (Mi et al., 2022; M(nT), Eq.
(9), at εr = 0.1 for interacting ( Randall et al., 2021).
Ji ≥ 0) and non-interacting (Ji = 0) chains.
However, quenched disorder is not The interacting chain shows period essential for realizing DTC order, and the doubling locked at 2T, while the non-asymptotic stability of MBL is itself under interacting chain shows beating.
(b) debate.
We therefore turn to clean Fourier spectrum of M(nT).
With periodically driven systems, where interactions the peak is pinned at ω = π/T; dynamical many body localization without interactions it shifts to ω = π (DMBL) can suppress heating through the intrinsic dynamics and provide a route (1−εr)/T.
toward time crystalline behavior in the The origin of this locking can be absence of disorder.
A representative understood from the structure of UF.
At εr example is a long-range Ising chain = 0, its eigenstates are cat states |{s}⟩ ± subject to a periodically modulated field, ϕ ei s|{−s}⟩ built from σz product states, y y = J σ σ with quasienergies differing by exactly π/T H2(t) ij i j i≤j .
A small rotation error flips individual z −hsin(Ωt) σi , (11) spins and therefore creates domain walls, i which cost an energy of order J .
When the J0 i J = α , ij |i−j| couplings and fields are random, these where Ω is the frequency of the field processes are generically off-resonant, so z modulation.
As demonstrated by Rahaman εr acts only perturbatively.
It dresses σi z et al.
(2024a,b), this model exhibits robust into quasi-local operators τi that still dynamical localization when the drive † z z satisfy UFτi UF ≃ −τi , and the π pairing parameters satisfy J0(4h/Ω) = 0, where J0 survives with corrections exponentially 9 denotes the zeroth-order Bessel function of τheat ≳ exp(c ωD /Jloc).
(12) the first kind.
The origin of this condition For times shorter than τheat, a suitable is clearest in the frame rotating with the rotating frame transformation brings the y y ± − field.
Writing σi σj in terms of σ , the flip-iDT Floquet operator to the form UF ≈ P e .
− + z flop terms σi σj + h.c.
conserve ∑i σi and The effective static Hamiltonian D has an are unaffected by the rotation.
The pair emergent Z2 symmetry, D,P = 0.
If the + + creation terms σi σj + h.c.

instead acquire, corresponding prethermal state up to a constant phase, the time dependent spontaneously breaks this symmetry, the phase e±i(4h/Ω)cosΩt.
Averaged over a discrete time translation symmetry is cycle, they are multiplied by J0(4h/Ω) and broken and the laboratory frame therefore vanish at its zeros.
To leading observable exhibits a robust period-order in 1/Ω, the effective Hamiltonian doubled response (Else et al., 2017).
Thus, then conserves the total magnetization, and the existence of a prethermal DTC is the resulting dynamical localization constrained by the same equilibrium suppresses heating (Rahaman et al., symmetry breaking considerations 2024a).
This localization mechanism can discussed above.
For short range be combined with the binary Floquet interactions, finite temperature breaking of a discrete symmetry requires d ≥ 2.
Clean protocol of Eq.
(5), with H2(t) acting long range interactions can modify this during the interval T2 (chosen as an conclusion and support time crystalline integer multiple of 2π/Ω), to stabilize a order even in one dimension (Pizzi et al., subharmonic response and thereby realize 2021; Russomanno et al., 2017).
A DTC order in a clean system (Rahaman et prethermal DTC of this kind has been al., 2024b).
observed in a trapped-ion chain A complementary route to realizing DTC (Kyprianidis et al., 2021).
order without disorder is provided by high-Another route to realizing DTC order in a frequency driving.
When the drive clean system is provided by a two-tone frequency ωD is much larger than the local drive protocol that dynamically cancels the energy scale Jloc, absorbing a single drive interaction contribution to the Floquet quantum requires a high-order many-body Hamiltonian, resulting in flat quasienergy process.
Consequently, the heating time is bands (Banerjee et al., 2025).
In this exponentially long (Abanin et al., 2017; scheme, the system is subjected to two Kuwahara et al., 2016; Mori et al., 2016), square wave drives with commensurate 10frequencies ω1 and ω2, satisfying ω1 = n protocol provides a route to realizing clean DTC order without relying on quenched ω2, where n ≥ 1 is an odd integer.
The corresponding time-dependent disorder.
Hamiltonian is III.
INTEGRABILITY AND THE (USUAL) ABSENCE OF = J σx σx H2(t) i i+1 i OSCILLATIONS + h [ sgn(sinω1t) (13) An integrable many-body system has an −sgn(sinω t) ] σz , 2 i extensive set of local or quasi-local i where sgn denotes the sign function.
For conserved charges {Qj} with H,Qj = the commensurate choice of drive Qj,Ql = 0 (Essler & Fagotti, 2016; frequencies specified above, the two-tone Ilievski et al., 2016).
Free-fermion models protocol cancels the interaction such as the transverse-field Ising chain contribution over a driving cycle.
The (Lieb et al., 1961; Pfeuty, 1970) belong to resulting flat quasienergy bands produce this class, as do interacting Bethe-ansatz-dynamical localization and thereby solvable models such as the XXZ and suppress heating.
When combined with a Hubbard chains.
Because these charges binary Floquet protocol that includes a constrain the dynamics, integrable systems spin flip operation, this localization do not thermalize.
After a quench they mechanism stabilizes a subharmonic relax locally to a generalized Gibbs response.
If the cancellation were exact, ensemble (GGE) (Ilievski et al., 2015; however, the evolution during T2 would Rigol et al., 2007; Vidmar & Rigol, 2016), act trivially and could not correct spin-rotation errors.
As in Fig.
2, the response 1 ρ = exp − β Q , (14) would then not be locked.
Small residual GGE Z j j j interactions restore this rigidity and where the Lagrange multipliers βj are fixed stabilize the response against spin-rotation by the initial values ⟨Q ⟩.
The quantum errors, which act as external perturbations.
j Rahaman and Roy (2026) showed that Newton’s cradle (Kinoshita et al., 2006) Floquet flat bands of this kind enable demonstrates this memory of initial discrete time crystalline order in spin conditions.
Periodically driven integrable chains, with a robust period doubled systems similarly relax to a periodic GGE stroboscopic magnetization.
Thus, this (Lazarides et al., 2014b).

11 Note that integrable systems do relax, just commensurability, and Sec.
III B discusses not to a thermal state.
Expand the state in its realization in free-fermion models.
energy eigenstates, |ψ⟩ = ∑ c |m⟩.
Then m m A.
Dynamical symmetries 2 The algebraic structure that produces ⟨O(t)⟩ = |cm| Omm m (15) exactly commensurate energy differences * i(Em−En)t + cm cn Omn e .
is a dynamical symmetry (Buča et al., m≠n 2019; Medenjak et al., 2020).
An operator In the thermodynamic limit, the energy A is a dynamical symmetry of a time-differences Em−En form a continuum.
independent Hamiltonian if * When the weights cmcnOmn are smoothly distributed over it, the second term H,A = ωA, (16)
dephases by the Riemann-Lebesgue with ω real.
It follows that A(t) = eiHtA lemma, leaving only the diagonal (GGE) − ω ω e iHt = ei tA, so that ⟨A(t)⟩ = ei t⟨A(0)⟩ part.
For a periodically driven system, the for every initial state, and the oscillation same argument applies at stroboscopic never decays.
In spectral terms, if H|E⟩ = times t = nT, with energies replaced by E|E⟩, then HA|E⟩ = (E + ω)A|E⟩.
The quasienergies.
Conserved charges with operator A therefore connects eigenstates finite overlap with an observable bound whose energies differ by exactly ω.
When the time average of its autocorrelation A is a sum of local terms, it supplies function from below (Mazur, 1969; precisely the extensive set of Suzuki, 1971), but this bound concerns the commensurate energy differences with stationary part of the response rather than finite weight that Eq.
(15) requires.
oscillations. Conservation laws therefore determine where the system relaxes, but The simplest example is the isotropic they do not by themselves prevent Heisenberg chain in a uniform field, H = J z relaxation.
Persistent oscillations require ∑i $\boldsymbol{S}$i · $\boldsymbol{S}$i+1 + h∑i Si .
Because the more: the spectrum must contain an exchange term is invariant under global extensive set of exactly commensurate spin rotations, the total raising operator energy differences, and these differences + + satisfies H,Stot = h Stot, and the total must contribute finite weight to local transverse magnetization precesses at observables.
Section III A explains what frequency h without decay.
This does not algebraic structure gives rise to such contradict the Watanabe-Oshikawa + theorem: ⟨Stot⟩ vanishes in every Gibbs 12 state of H, so the oscillation requires a structure of the drive, P2 = $\mathbf{1}$.
As long as nonequilibrium initial state.
A second weak perturbations preserve the dressed − example is Yang's η-pairing operator in iDT form UF ≈ P e with D,P = 0, the the Hubbard model on a bipartite lattice, phase θ = π cannot drift continuously.
This which satisfies Eq.
(16) with a frequency is the algebraic origin of the rigidity of the set by the on-site interaction and the period in a discrete time crystal, and it chemical potential (Yang, 1989).
distinguishes Floquet dynamical Medenjak et al.
(2020) argued that isolated symmetries from their Hamiltonian Heisenberg magnets with such structure counterparts.
Floquet dynamical can be regarded as quantum time crystals.
symmetries of this type have been used to Long-lived phantom-helix states, whose construct time crystals in driven Hubbard longevity reflects exact properties of the models (Chinzei & Ikeda, 2020).
The free-anisotropic Heisenberg model, have been fermion models of Sec.
III B provide a observed with ultracold atoms (Jepsen et further example, in which the condition is al., 2022).
In all these cases, however, the met for selected momenta.
frequency is set by Hamiltonian B.
Free-fermion models parameters and shifts continuously with them.
Such oscillations are therefore not Time-crystalline responses can be realized rigid in the sense of criterion (b).
in a class of integrable free-fermion models (Chandra & Roy, 2024), For a periodically driven system, the analogue of Eq.

(16) is (0) H(t) = H$\boldsymbol{k}$ , θ $\boldsymbol{k}$≥0 U† A U = ei A, (17) (18) F F (0) = $\boldsymbol{k}$ $\boldsymbol{k}$ − θ H$\boldsymbol{k}$ (a0−b )(n + n $\boldsymbol{k}$) in which gives A(nT) = e A.
A phase θ = 2 † †  +D c$\boldsymbol{k}$c $\boldsymbol{k}$ + h.c.
, $\boldsymbol{k}$ − π/m produces a response with period mT, † with n±$\boldsymbol{k}$ = c±$\boldsymbol{k}$c±$\boldsymbol{k}$.
This Cooper-pair and θ = π gives period doubling.
In Hamiltonian is specified by the parameters spectral terms, A connects Floquet a0, b , and D and by the fermionic ladder eigenstates whose quasienergies differ by $\boldsymbol{k}$ $\boldsymbol{k}$ † operators c±$\boldsymbol{k}$ and c±$\boldsymbol{k}$.
It consists of a θ/T, which for θ = π is the π pairing of kinetic term with coupling a −b and a 0 $\boldsymbol{k}$ Sec.
II B.
In the kicked Ising chain at εr = pairing term with coupling D , where a is z $\boldsymbol{k}$ 0 0, every σi satisfies Eq.
(17) with θ = π.
time dependent.
Consider driving the Here the phase is not set by a tunable system with a square-wave signal of Hamiltonian parameter but by the Z2 period T and a 50% duty cycle, such that 13 the first half-cycle flips the spins while the a next-nearest-neighbor (NNN), or second half-cycle does not.
The equivalently three-spin, interaction.
The Hamiltonian during the nth cycle is then model remains integrable, while the (0) subharmonic conditions become a0 = bk H$\boldsymbol{k}$ , 1 = cosk + λcos2k and ωD = 2sink + 2λsin  nT ≤ t ≤ n T, + 2 H (t) = † † (19) 2k, where λ denotes the three-spin $\boldsymbol{k}$ a1 c$\boldsymbol{k}$c−$\boldsymbol{k}$ + h.c.
, 1 coupling.
This additional interaction  n T ≤ t ≤ n 1 T.
+ 2 ( + ) provides the tunability needed to construct For a particular momentum, $\boldsymbol{k}$ = $\boldsymbol{k}$ , if a 0 0 a 1D time crystal and overcome the = b and ω ≡ 2π/T = 2D , the wave $\boldsymbol{k}$0 D $\boldsymbol{k}$0 limitations of the 1D TFIM.
function repeats after a time 2T.
This subharmonic response is a hallmark of IV.
EXPERIMENTAL discrete time-crystalline behavior.
REALIZATIONS For the one-dimensional (1D) transverse-The first observations of discrete time field Ising model (TFIM), the subharmonic crystalline order used two complementary platforms.
Zhang et al.
(2017) drove a conditions reduce to and a0 = bk0 = cosk0 chain of trapped ions with long range Ising ωD = 2sink0.
However, a small deviation interactions and engineered disorder.
They of the momentum from the specific value observed a period doubled magnetization k0 destroys the time-crystalline response.
that remained locked at 2T over a finite In the two-dimensional (2D) TFIM, by range of pulse imperfections and melted contrast, the conditions become a = b = 0 $\boldsymbol{k}$ when the imperfections became large.
cosk + αcosk and ω = 2sink + 2αsin x y D x Because the chain was finite and the ky.
Setting α = 0 recovers the 1D case.
interactions long ranged, it is difficult to The additional tunable parameter in two establish that the stabilization in this dimensions therefore permits a continuous experiment is genuinely due to many body range of momenta supporting a stable time localization.
Choi et al.
(2017) used a crystalline response.
In the 1D TFIM, no dense ensemble of nitrogen-vacancy (NV) analogous parameter is available, centers in diamond with disordered dipolar precluding the construction of a stable interactions.
In this system the order is not time crystal by this mechanism.
stabilized by localization; instead, slow, To obtain stable time-crystalline responses critical thermalization of the dipolar in 1D, one can supplement the TFIM with ensemble extends its lifetime.
Both 14 experiments demonstrated a rigid al.
(2022) used a superconducting quantum subharmonic response, but neither could processor to show that time-crystalline test persistence to infinite times.

order persists across the spectrum, Experiments in clean systems followed, including for random initial states.
Their which is the setting of primary interest in echo protocol was designed to separate the this chapter.
NMR experiments on star intrinsic decay of the order from shaped molecular clusters (Pal et al., 2018) decoherence.
A realization on a large and on a spatially ordered dipolar crystal cloud based quantum computer followed (Rovny et al., 2018) showed signatures of (Frey & Rachel, 2022), showing that such Floquet circuits can now be implemented time crystalline order without engineered on general purpose quantum hardware.
disorder. The mechanism that stabilizes the response in the dipolar crystal is still Time crystalline behaviour has also been debated, with prethermalization and long observed beyond closed spin systems.
range interactions as the main candidates.
Magnon condensates in superfluid 3He A prethermal discrete time crystal was exhibit a time quasicrystal and its subsequently demonstrated with trapped transition to a time crystal (Autti et al., ions (Kyprianidis et al., 2021).
Its lifetime 2018).
In atom cavity systems, dissipation grows with the drive frequency, as stabilizes a period doubled response expected from the slow heating of the (Keßler et al., 2021), and continuous prethermal regime discussed in Sec.
II B.
pumping produces self sustained In Rydberg atom arrays (Bluvstein et al., oscillations that break continuous time 2021), periodic driving stabilizes period translation symmetry (Kongkhambut et al., doubled revivals of quantum many-body 2022).
These open system realizations scarred states (Serbyn et al., 2021; Turner show that dissipation can stabilize, rather et al., 2018).
This experiment shows that than only destroy, time crystalline order.
subharmonic dynamics can arise from Several of these experiments connect restricted, rather than complete, algebraic directly to the themes of this chapter.
structure in the spectrum.
Floquet circuits on superconducting Later experiments addressed eigenstate processors closely match the kicked Ising order directly.
Randall et al.
(2021) used a chain of Eqs.
(5) and (6), with integrability programmable register of 13C nuclear spins breaking terms that can be tuned.
When surrounding an NV center to probe a many the longitudinal fields vanish, the circuit body localized discrete time crystal.
Mi et reduces to a free fermion model, so the 15 free fermion physics discussed in Sec.
III The first concerns clean interacting B can be probed directly.
The phantom systems.
It is not known whether there helix experiments in ultracold atoms exists a rigorous, non perturbative example (Jepsen et al., 2022) probe the dynamical of a bulk discrete time crystal in a clean, symmetry physics of Sec.
III A in an short range, one dimensional system that isolated Heisenberg magnet.
In all persists indefinitely.
Prethermal platforms, however, the observation time mechanisms cannot provide such an is limited by decoherence, and hybrid example, since a discrete symmetry cannot quantum-classical feedback has been be spontaneously broken at finite proposed as a way to prolong discrete time temperature in one dimension with short crystalline order on such hardware range interactions.
Edge modes offer an (Camacho & Fauseweh, 2024).
Separating alternative at the boundary.
Floquet intrinsic persistence, as required by Majorana π modes produce persistent criterion (c), from experimental limitations period doubling at the ends of driven free therefore remains a central challenge.
fermion chains (Thakurathi et al., 2013), and strong and almost strong zero modes V.
OPEN PROBLEMS AND give long lived edge spins in interacting OUTLOOK chains (Fendley, 2016; Kemp et al., 2017).
Integrable models give an exact, Extending such edge modes to interacting analytically controlled view of what a time Floquet chains provides one route toward crystal requires.
They show that persistent long lived boundary oscillations.
In the oscillations need commensurate spectral bulk, clean long range interactions in one structure, expressed through dynamical

dimension (Pizzi et al., 2021; Russomanno symmetries and the π pairing of Floquet et al., 2017), or short range interactions in eigenstates (Secs.
II B and III A).
In the higher dimensions (Else et al., 2017), one-dimensional TFIM this structure is allow time crystalline order without confined to isolated momenta (Sec.
III B).
disorder. Interactions are needed to lock the A second direction concerns the robustness response and to stabilize it against the of dynamical symmetries.
The dynamical domain walls created by pulse errors (Sec.
symmetries of Heisenberg and Hubbard II B).
Integrable models thus explain why models discussed in Sec.
III A produce the structure required for a time crystal is persistent oscillations, but their generically fragile.
Several directions frequencies are set by Hamiltonian remain open.
16 parameters and shift continuously with 2022).
Algebraic conditions for persistent them.
They are therefore not rigid in the oscillations and limit cycles under sense of criterion (b).
An important dissipation have been formulated (Buča et question is whether such symmetries can al., 2022, 2019).
How these open system be made robust, for instance through drive-phases relate to the closed system, induced locking, so that their frequencies eigenstate based definition of a discrete become rigid.
The contrast with the Z2 time crystal, and to classical discrete time structure of the kicked Ising chain suggests crystals (Zaletel et al., 2023), remains to that a discrete symmetry of the drive is the be clarified.
essential ingredient.
Responses beyond period doubling also A third direction lies between the deserve attention.
Higher-order and integrable and chaotic limits.
Quantum fractional subharmonic responses (Pizzi et many body scars (Serbyn et al., 2021; al., 2021), as well as time quasicrystals Turner et al., 2018) and Hilbert space (Autti et al., 2018), remain largely fragmentation (Sala et al., 2020) generate unexplored in integrable settings, where long lived oscillations from restricted exact solutions could clarify their spectral algebraic structure.
In scarred systems, origin.
Equation (17) with θ = 2π/m approximately equally spaced towers of provides a natural starting point for such eigenstates play the role of a dynamical an analysis.
symmetry acting within a subspace, and Finally, the asymptotic status of many periodic driving can stabilize the resulting body localization remains unsettled.
revivals (Bluvstein et al., 2021).
This Although MBL has been proven in one similarity suggests that a single algebraic dimension under an assumption on level framework may describe non stationary statistics (Imbrie, 2016), numerical studies many body dynamics.
questioning its stability (Šuntajs et al., Open systems raise a further set of 2020) and the identification of avalanche questions.
Dissipation can stabilize time instabilities (De Roeck & Huveneers, crystalline order, as in boundary time 2017; Morningstar et al., 2022) have crystals (Iemini et al., 2018), in dissipative reopened the question.
Its resolution will discrete time crystals without manifest determine whether disordered discrete symmetries or disorder (Gambetta et al., time crystals are true phases of matter or 2019), and in driven atom cavity systems merely extremely long lived crossovers.
(Keßler et al., 2021; Kongkhambut et al., This question also motivates the disorder 17 free mechanisms emphasized in this for periodically driven and closed review.
quantum systems.
Communications in Mathematical Physics, 354(3), 809– Time crystals extend the concept of spontaneous symmetry breaking to 827.
https://doi.org/10.1007/s00220-systems far from equilibrium.
Their 017-2930-x defining feature is not oscillation itself but Abanin, D.
A., Altman, E., Bloch, I., & its rigidity, which in Floquet systems Serbyn, M.

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fraction of the drive frequency.
Integrable Reviews of Modern Physics, 91, models show most clearly what this 021001.
requires, namely commensurate spectral https://doi.org/10.1103/RevModPhys.9 structure protected against perturbations, 1.021001 and why generic systems struggle to Autti, S., Eltsov, V.
B., & Volovik, G.
E. provide it.
Whether clean interacting (2018).
Observation of a time systems can provide it indefinitely, rather quasicrystal and its transition to a than for exponentially long but finite superfluid time crystal.
Physical times, remains the central open question of Review Letters, 120, 215301.
the field.
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